



Maximum Marks: 100		Semester: July 2024 - November 2024	
Examination: ESE Examination		Duration: 3 Hrs.	
Programme code: 03		Class: SY	Semester: III (SVU 2020)
Programme: B. Tech Electronics and Telecommunication Engineering			
Name of the Constituent College: K. J. Somaiya School of Engineering		Name of the department: Electronics and Telecommunication Engineering	
Course Code: 116U03C301	Name of the Course: Mathematics For Communication Engineering-I		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any Four of the following.	20
i)	Find $L(te^{-4t} \sin 3t)$	5
ii)	Find $L^{-1} \left[\frac{2s+3}{s^2+2s+2} \right]$	5
iii)	Find Fourier Coefficient a_n for the function $f(x) = \begin{cases} x + \frac{\pi}{2}, & -\pi < x < 0 \\ \frac{\pi}{2} - x, & 0 < x < \pi \end{cases}$	5
iv)	Express $J_4(x)$ in terms of $J_0(x)$ and $J_1(x)$.	5
v)	If $\vec{f} = x^2\vec{i} + xz\vec{j} + yz\vec{k}$ and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, find $\text{div}(\vec{f} \times \vec{r})$	5
vi)	Evaluate $\int_A^B 3xydx - y^2dy$ along the straight line joining $A(0,0)$ and $B(1,2)$.	5
Q2 A	Solve the following.	10
i)	If $L\{f(t)\} = \frac{e^{-1/s}}{s}$, then find $L\{e^{-2t}f(3t)\}$	5
ii)	Find Inverse Laplace transform of $\frac{1}{(s-1)(s^2+4)}$ using Convolution theorem	5
OR		
Q2 A	Find the Fourier series for $f(x) = 1 - x^2$ when $-1 \leq x \leq 1$.	10
Q 2 B	Solve any One of the following.	10
i)	$\frac{d^2y}{dt^2} + 4y = 1$ with conditions $y(0) = 0, y'(0) = 1$	10
ii)	Obtain the Fourier series expansion of $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in the interval $0 \leq x \leq 2\pi$, and $f(x+2\pi) = f(x)$. Also deduce that $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$	10

Q3	Solve any Two of the following.	20
i)	(a) Obtain Complex form of Fourier series for $f(x) = e^{5x}$ in $(-5, 5)$	05
	(b) Prove that $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$	05
ii)	Find the directional derivative of $\phi = xyz$ in the direction normal to the surface $x^2y + y^2x + yz^2 = 3$ at the point $(1, 1, 1)$.	10
iii)	Show that $\vec{F} = (ye^{xy} \cos z)i + (xe^{xy} \cos z)j - (e^{xy} \sin z)k$ is irrotational and find the scalar potential for $\vec{F}$ and evaluate $\int \vec{F} \cdot d\vec{r}$ along the curve joining the points $(0, 0, 0)$ and $(-1, 2, \pi)$	10
Q4	Solve any Two of the following.	20
i)	(a) Obtain b_n for half-range sine series of $f(x) = \begin{cases} x, & 0 < x < \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} < x < \pi \end{cases}$	5
	(b) Prove that $\int x^4 J_1(x) dx = x^4 J_2(x) - 2x^3 J_3(x)$	5
ii)	Define divergence and gradient and prove that $\text{div}(\text{grad}(r^n)) = n(n+1)r^{n-2}$ where $r = \vec{r} $	10
iii)	Verify Green's theorem for $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ along C, where C is the boundary of the region bounded by $x = y^2$ and $y = x$.	10
Q5	Solve any Four of the following.	20
i)	Using Laplace Transform evaluate $\int_0^\infty e^{-t}(1 + 2t - 3t^2) dt$	5
ii)	Find $L^{-1} \left[\tan^{-1} \left(\frac{2}{s} \right) \right]$	5
iii)	Obtain half-range cosine series for $f(x) = x$ for $0 \leq x \leq 1$	5
iv)	If $\vec{f} = (x + y + 1)i + j - (x + y)k$, prove that $\vec{f} \cdot \text{Curl } \vec{f} = 0$	5
v)	Prove that $(\vec{a} \times \vec{b}) \cdot [(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})] = [\vec{a} \cdot (\vec{b} \times \vec{c})]^2$	5
vi)	Use Gauss's Divergence theorem to evaluate $\iint \vec{N} \cdot \vec{F} ds$, where $\vec{F} = 2xi + xyj + zk$, over the region bounded by the cylinder $x^2 + y^2 = 4, z = 0, z = 6$	5



Semester: July 2024-November 2024			Duration: 2 Hrs.
Maximum Marks: 50	Examination: End-Semester Examination		
Programme code: 03	Class: SY	Semester: III (SVU 2023)	
Programme: B. Tech Electronics and Telecommunication Engineering	Name of the department: Electronics and Telecommunication Engineering		
Name of the College: K. J. Somaiya School of Engineering	Name of the Course: Mathematics For Communication Engineering		
Course Code: 216U03C301			
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	
i)	If $\vec{a} = 3\hat{i} - 2\hat{j} + 2\hat{k}$, $\vec{b} = 6\hat{i} + 4\hat{j} - 2\hat{k}$, $\vec{c} = 3\hat{i} + 2\hat{j} + 4\hat{k}$, find $\vec{a} \times (\vec{b} \times \vec{c})$	05
ii)	Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = 2xi + (xz - y)j + 2zk$ from $O(0, 0, 0)$ to $P(3, 1, 2)$ along the line OP	05
iii)	Find the Fourier series for $f(x)$, if $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ \pi & 0 < x < \pi \end{cases}$	05
Q.2	Attempt any one	10
i)	(a) Obtain a half-range cosine series for $f(x) = x + 1$ in $(0, 2)$ (b) Obtain Complex form of Fourier series of $f(x) = \cosh x + \sinh x$ in $(-\pi, \pi)$	
ii)	If $\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational then (a) Find the constants a, b, c (b) Find scalar potential for $\vec{F}$ (c) Find the work done in moving a particle in field $\vec{F}$ from $(1, 2, -4)$ to $(3, 3, 2)$ along the straight line joining these points	



Q.3	Attempt any one	10
i)	Define Bessel's Function and obtain $J_{1/2}(x)$ and $J_{-1/2}(x)$ in terms of $\sin x$ and $\cos x$ respectively. Hence prove that $J_{1/2}^2(x) + J_{-1/2}^2(x) = \frac{2}{\pi x}$	
ii)	(a) Find $L\left(\frac{\cos h 2t \sin 2t}{t}\right)$ (b) Find $L^{-1}\left(\frac{s+4}{s(s^2-4)}\right)$ using partial fraction method.	
Q.4	Attempt the following	10
	Verify Green's theorem for $\vec{F} = xy\hat{i} + x^2\hat{j}$, and C is the triangle having vertices at $(0, 0)$, $(1, 1)$ and $(1, -1)$	
Q.5	Attempt any two	
i)	Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $P(2, -1, 2)$ in the direction from this point towards the point $Q(4, -4, 8)$.	05
ii)	By using Stoke's Theorem evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = xy\hat{i} + xy^2\hat{j}$ and C is the square in the xy -plane with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$ and $(0, -1)$	05
iii)	Evaluate $\int_0^\infty e^{-3t} t \cos t dt$ using Laplace transform.	05



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18.11.2024(E)

Semester: July 2024 – November 2024		
Maximum Marks: 50	Examination: ESE Examination (PWD SECTION – A)	Duration: 2 Hrs.
Programme code: 03		
Programme: B. Tech Electronics and Telecommunication Engineering	Class: SY	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya School of Engineering	Name of the department: Electronics and Telecommunication Engineering	
Course Code: 116U03C301	Name of the Course: Mathematics For Communication Engineering-I	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que No.	Question	Marks
Q.1	Attempt any FOUR of the following	
(i)	Find $L(te^{-4t} \sin 3t)$	05
(ii)	Find $L^{-1} \left[\frac{2s+3}{s^2+2s+2} \right]$	05
(iii)	If $L\{f(t)\} = \frac{e^{-1/s}}{s}$, then find $L\{e^{-2t}f(3t)\}$	05
(iv)	Find Fourier Coefficient a_n for the function $f(x) = \begin{cases} x + \frac{\pi}{2}, & -\pi < x < 0 \\ \frac{\pi}{2} - x, & 0 < x < \pi \end{cases}$	05
(v)	Obtain half-range cosine series for $f(x) = x$ for $0 \leq x \leq 1$	05
(vi)	Obtain Fourier Coefficient c_n of Complex form of Fourier series for $f(x) = e^{5x}$ in $(-5, 5)$	05
Q.2	Attempt any FOUR of the following	
(i)	Find Inverse Laplace transform of $\frac{1}{(s-1)(s^2+4)}$ using Convolution theorem	05
(ii)	Using Laplace Transform evaluate $\int_0^\infty e^{-t}(1+2t-3t^2) dt$.	05
(iii)	Find $L^{-1} \left[\tan^{-1} \left(\frac{2}{s} \right) \right]$	05
(iv)	Obtain the Fourier Coefficient b_n for $f(x) = \left(\frac{\pi-x}{2} \right)^2$ in the interval $0 \leq x \leq 2\pi$, and $f(x+2\pi) = f(x)$.	05
(v)	Find the Fourier coefficient a_0 for $f(x) = 2 - x^2$ in the interval $(0, 2)$.	05

(vi)	If $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ E \sin x, & 0 < x < \pi \end{cases}$ then find the value of Fourier coefficient b_1 for $f(x)$	05
Q.3	Attempt the following	
(i)	If $f(t) = \sin t \cos t$, then $L[f(t)] =$ (a) $\frac{s}{s^2+2^2}$ (b) $\frac{2}{s^2+2^2}$ (c) $\frac{1}{2} \left[\frac{1}{s^2+2^2} \right]$ (d) $\frac{1}{s^2+2^2}$	02
(ii)	Evaluate $\int_0^\infty \frac{\sin 3t}{t} dt$ (a) $\frac{1}{s} \left(\frac{\pi}{2} - \tan^{-1} \frac{s}{3} \right)$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2} - \tan^{-1} \frac{s}{3}$ (d) $\frac{\pi}{2}$	02
(iii)	Laplace transform of $t^4(\sinh at + \cosh at)^n$ is (a) $\frac{24}{(s-an)^5}$ (b) $\frac{24}{(s+an)^5}$ (c) $\frac{24}{(s-a)^5}$ (d) $\frac{24}{(s+a)^5}$	02
(iv)	If for $f(x) = \frac{1}{2}(\pi - x)$ in $(0, 2\pi)$, $f(x) = \sum_{n=1}^\infty \frac{1}{n} \sin nx$ then $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots =$ (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $-\frac{\pi}{4}$ (d) $-\frac{\pi}{2}$	02
(v)	If $x \sin x = -1 - \frac{1}{2} \cos x + \pi \sin x + \sum_{n=2}^\infty \frac{2 \cos nx}{n^2-1}$ in $(0, 2\pi)$ For which value of x we get the following series $\sum_{n=2}^\infty \frac{1}{n^2-1} = \frac{3}{4}$ (a) 0 (b) π (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$	02



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EXR ETR

18.11.2024 (E)

Maximum Marks: 50		Semester: July 2024 – November 2024		Duration: 2 Hrs.
		Examination: ESE Examination (PWD SECTION – B)		
Programme code: 03		Class: SY	Semester: III (SVU 2020)	
Programme: B. Tech Electronics and Telecommunication Engineering				
Name of the Constituent College: K. J. Somaiya School of Engineering		Name of the department: Electronics and Telecommunication Engineering		
Course Code: 116U03C301	Name of the Course: Mathematics For Communication Engineering-I			
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary				

Que No.	Question	Marks
Q.4	Attempt any FOUR of the following	
(i)	If $\vec{f} = x^2\vec{i} + xz\vec{j} + yz\vec{k}$ and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, find $\text{div}(\vec{f} \times \vec{r})$	05
(ii)	If $\vec{f} = (x + y + 1)\vec{i} + \vec{j} - (x + y)\vec{k}$, prove that $\vec{f} \cdot \text{Curl } \vec{f} = 0$	05
(iii)	Show that $\vec{F} = (ye^{xy} \cos z)\vec{i} + (xe^{xy} \cos z)\vec{j} - (e^{xy} \sin z)\vec{k}$ is irrotational and Find the scalar potential for $\vec{F}$.	05
(iv)	Evaluate $\int_A^B 3xydx - y^2dy$ along the straight line joining $A(0, 0)$ and $B(1, 2)$	05
(v)	Use Green's theorem to evaluate $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ along C, where C is the boundary of the region bounded by $x = y^2$ and $y = x$.	05
(vi)	Use Gauss's Divergence theorem to evaluate $\iint \vec{N} \cdot \vec{F} ds$, where $\vec{F} = 2x\vec{i} + xy\vec{j} + z\vec{k}$ over the region bounded by the cylinder $x^2 + y^2 = 4$, $z = 0$, $z = 6$	05
Q.5	Attempt any FOUR of the following	
(i)	Prove that $(\vec{a} \times \vec{b}) \cdot [(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})] = [\vec{a} \cdot (\vec{b} \times \vec{c})]^2$	05
(ii)	Find the maximum directional derivative of $\phi = 4x - y + 2z$ at $(1, 2, 1)$.	05
(iii)	If $\vec{F} = (x + 3y)\vec{i} + (y - 2z)\vec{j} + (az + x)\vec{k}$ is solenoidal, Find the value of a.	05
(iv)	Find the unit normal to the surface $xy^3z^2 = 4$ at $(-1, -1, 2)$.	05

(v)	A vector field $\vec{F}$ is given by $\vec{F} = (\sin y)i + x(1 + \cos y)j$. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where C is the closed surface of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, z = 0$.	05
(vi)	Express $J_4(x)$ in terms of $J_0(x)$ and $J_1(x)$.	05
Q.6	Attempt the following	
(i)	$\nabla\phi$ represents geometrically (a) tangent to the surface $\phi(x, y, z) = c$ (b) Normal to the surface $\phi(x, y, z) = c$ (c) derivative of ϕ in direction of $\nabla\phi$ (d) Direction Derivative of ϕ	02
(ii)	If $\vec{A} = \nabla(xy + yz + zx)$ then $\nabla \cdot \vec{A}$ is (a) 3 (b) 0 (c) $xi + yj + zk$ (d) $x + y + z$	02
(iii)	Let $\phi = xy + yz + zx$ Find the rate of change of ϕ in the direction of the normal to the surface $x^2 + y^2 = z + 4$ at $(1, 1, -2)$. (a) -2 (b) 2 (c) 0 (d) -6	02
(iv)	Gauss divergence theorem states that (a) $\iint_S \vec{N} \times \vec{F} ds = \iiint_V \nabla \times \vec{F} dv$ (b) $\iint_S \vec{N} \cdot \vec{F} ds = \iiint_V \nabla \cdot \vec{F} dv$ (c) $\iint_S \vec{N} \cdot \vec{F} ds = \iiint_V \nabla \times \vec{F} dv$ (d) $\iint_S \vec{N} \times \vec{F} ds = \iiint_V \nabla \cdot \vec{F} dv$	02
(v)	Which of the property is correct for $J_n(x)$ (a) $\frac{d}{dx}(x^n J_n(x)) = x^n J_{n-1}(x)$ (b) $\frac{d}{dx}(x^{n+1} J_n(x)) = x^n J_{n-1}(x)$ (c) $\frac{d}{dx}(x^n J_n(x)) = x^n J_{n+1}(x)$ (d) $\frac{d}{dx}(x^n J_{n-1}(x)) = x^n J_{n-1}(x)$	02

22-11-2024(E)


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Semester: July 2024 –November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: __03	Class: SY	Semester:III_(SVU 2020)
Programme: __BTech EXTC		
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 116U03C303	Name of the Course: Digital Logic Circuits	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Evaluate 1. $(78)_{10} - (53)_{10}$ using 2's Complement method 2. $(328)_{\text{Hex}} + (16B)_{\text{Hex}}$	5
ii)	Draw JK flip flop using gates. Write truth table for the same.	5
iii)	Realise 2input EX-OR gate using NAND gates only	5
iv)	What is parity? How parity is used for error detection?	5
v)	Write truth tables of a 3 bit ring counter and 3 bit twisted ring counter.	5
vi)	What is a decade counter? Write truth table of the decade counter	5

Q2 A	Solve the following	10
i)	Prove the expression $(AB + C + D)(\bar{C} + D)(\bar{C} + D + E) = ABC + D$	5
ii)	Perform following conversions 1. $(10101010)_2 = (\text{grey code})$ 2. $(324)_{16} = (?)_{\text{BCD}}$	5
	OR	
Q2 A	Use Quine – Mc Cluskey to minimize the following expression $F(ABCD) = \sum m(0, 2, 4, 6, 7, 8, 9, 10, 13)$	10
Q 2 B	Solve any One	10
i)	Design and implement a Full Adder using suitable decoder. Assume that output of the decoder is active low.	10
ii)	Realize the following function using only one 8:1 Multiplexer. $Y(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 11, 13, 15)$.	10

Q3	Solve any Two	20
i)	Draw 4 bit Johnson/ Twisted ring counter using D flip-flops with negative edge triggering. Write truth table and draw timing diagram. (Assume: all flip-flops are cleared initially)	10
ii)	Explain the Race around condition in JK flip flop with circuit diagram and necessary waveform. How the same is overcome?	10
iii)	Design a synchronous counter that goes from 0 to 6 and resets back to 0 using JK flip flop.	10

Que. No.	Question	Max. Marks																										
Q4	Solve any Two	20																										
i)	Using any method of state reduction, reduce following state table/PS NS table. Draw reduced state diagram <table border="1"> <thead> <tr> <th rowspan="2">Present state(PS)</th><th colspan="2">Next state (NS) / Output</th></tr> <tr> <th>X=0</th><th>X=1</th></tr> </thead> <tbody> <tr><td>A</td><td>D/0</td><td>B/0</td></tr> <tr><td>B</td><td>E/0</td><td>A/1</td></tr> <tr><td>C</td><td>G/0</td><td>F/1</td></tr> <tr><td>D</td><td>A/1</td><td>D/0</td></tr> <tr><td>E</td><td>A/1</td><td>D/0</td></tr> <tr><td>F</td><td>C/0</td><td>B/0</td></tr> <tr><td>G</td><td>A/1</td><td>E/0</td></tr> </tbody> </table>	Present state(PS)	Next state (NS) / Output		X=0	X=1	A	D/0	B/0	B	E/0	A/1	C	G/0	F/1	D	A/1	D/0	E	A/1	D/0	F	C/0	B/0	G	A/1	E/0	10
Present state(PS)	Next state (NS) / Output																											
	X=0	X=1																										
A	D/0	B/0																										
B	E/0	A/1																										
C	G/0	F/1																										
D	A/1	D/0																										
E	A/1	D/0																										
F	C/0	B/0																										
G	A/1	E/0																										
ii)	Draw circuit diagram of a 3 bit left shift register in SIPO mode. Explain working of the same with an example.																											
iii)	Design Asynchronous Mod 11 up counter using toggle (T) flip flops and additional gates if required. Assume negative edge triggered flip flops.	10																										

Q5	Solve any four	20
i)	Distinguish Between synchronous and asynchronous counters (At least 5 points)	5
ii)	What is Canonical form of SOP? Explain with an example	5
iii)	State D'Morgans laws and their application	5
iv)	What is Radix or Base of a number system? Explain with example.	5
v)	Explain what is Moore and Melay machine with block diagram.	5
vi)	Simplify the expression using Boolean Laws $AB(\bar{B} + C) + BC + B$	5



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22.11.2024(E)

Semester: July 2024– November 2024		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs.
Programme code: 03 Programme: BTech EXTC	Class: SY BTech EXTC	Semester: III (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department:EXTC	
Course Code: 216U03C303	Name of the Course: Digital Logic Design	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	
i)	What is magnitude comparator? Draw truth table of 2 bit magnitude comparator. (Assume number A_1A_0 is compared with B_1B_0)	05
ii)	Draw and Explain Successive approximation technique of Analog to Digital Conversion	05
iii)	Realise following functions using one 3 to 8 line decoder and gates. Assume decoder has active low output. $F1 = \sum m(0, 2, 4, 5)$, $F2 = \sum m(0, 3, 4, 7)$	05
Q.2	Attempt any one	10
i)	a) If $Y(A,B,C,D) = ABC + D$ Draw the corresponding K map representing above expression b) $F = \sum m(0, 2, 4, 5, 6, 7, 10, 11, 14, 15)$. Realise using only one 8:1 mux	5 5
	OR	
i)	Reduce the following expression using Quine-McClusky method: $F(A,B,C,D) = \sum m(0, 1, 3, 9, 11, 12, 14, 15)$	10
Q.3	Attempt any one	10
i)	Write truth tables and Excitation tables of S-R, J-K and D flip flops? Explain what is race around condition in flip flop? How the same can be overcome?	10
ii)	Draw and explain 4 bit Johnson's Ring Counter with neat diagram and truth table. Draw Timing diagram.	10
Q.4	Attempt the following Design a mod 6 synchronous counter using JK flip flops	10
Q.5	Attempt the following	10
A)	Perform Following Conversions (Show steps of conversions) 1. Convert $(A20)_{16}$ into decimal. 2. $(543)_{10}$ into BCD	5
	OR	
A)	Prove the expression $(AB + C + D)(\bar{C} + D)(\bar{C} + D + E) = AB\bar{C} + D$	
B)	Differentiate between PROM, PAL and PLA	5



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2110

Semester: July 2024 – November 2024		
Maximum Marks: 50 (Section A)		
Examination: ESE Examination		Duration: 2 Hrs.
Programme code: 03		Class: SY
Programme: BTech EXTC		BTech EXTC
Semester: III (SVU 2020)		
Institute/School/Department: K. J. Somaiya School of Engineering		Name of the department: EXTC
Course Code: 116U03C303	Name of the Course: Digital Logic Design (PWD Section A)	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Implement AND , OR gates using NOR gates	5
ii)	What is magnitude comparator? If two numbers A and B are compared, write truth table considering all possibilities of inputs and outputs.	5
iii)	State De Morgan's Theorems and explain with diagrams	5
iv)	Perform following conversions a) Decimal (95) to Binary b) Hex (34B) to Binary and Decimal	5
v)	Perform following Arithmetic Operations Decimal (46) - Decimal (21) using 2's complement subtraction Hint: (Add 2's complement of 21 to 46)	5
vi)	Write short note on weighted and Non- weighted Codes	5

Que. No.	Question	Max. Marks
Q2	Solve the following (Any 2)	20
i)	If $F(A,B,C,D) = \sum m(0,1,3,9,11,12,14,15)$ a) Draw 4 variable K map of the function. b) Draw the circuit using gates	10
ii)	a) What is decoder? Draw block diagram . b) Realise following function using one 3 to 8 line decoder and gates . $F1 = \sum m(0, 2, 4, 5)$	10
iii)	Draw circuit diagram of controlled adder / subtractor using 4 bit adder IC and explain with an example	10
iv)	Design a full adder using logic gates . a) Write truth table indicating sum and carry output b) Draw K maps for sum and carry c) Find simplified equations d) Draw logic diagram	10

Que. No.	Questions	Max. Marks
Q3	Solve following Multiple choice Questions	10
i)	The number of digits in octal systems are A. 8 B. 7 C. 18 D. 10	1
ii)	Decimal number 9 in Gray code is A. 1100 B. 1101 C. 0110 D. 1111	1
iii)	The number FF in hexadecimal system has equivalence in decimal system to A. 265 B. 255 C. 265 D. 239	1
iv)	With reference to adders Which of the following statement is NOT correct? A. A full adder can be constructed using two half adders and an OR gate B. Two four bit parallel adders can be cascaded to construct 8 bit parallel adder C. Total Ripple carry addition time is independent of number of bits to be added D. Carry look ahead is used to speed up parallel addition	1
v)	1. Which of the following expression represents a D Morgan's Theorem identity? A. $1+A=1$ B. $A(B+C)=AB+AC$ C. $\overline{(A+B)} = \overline{A} \cdot \overline{B}$ D. $A + B \overline{A} = A+B$	1
vi)	-24 in 2's complement form is A. 11101000 B. 01001000 C. 01111111 D. 00111111	2
vii)	If 32:1 multiplexer is to be designed using only 4:1 multiplexer ICs , number of ICs required for realization will be A. 8 B. 11 C. 12 D. 13	2
viii)	Which of the following statement is NOT true for a programmable logic Device (PLD)? A. PLDs are complex than combinational circuits B. PLDs have programmable fuses C. Programmable Logic arrays (PLA) has more flexibility than Programmable Read only Memory (PROM). D. PLDs have arrays of NAND and EXOR gates	1

22.11.2024 (E)

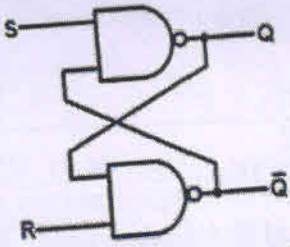

SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: July 2024 – November 2024		
Maximum Marks: 50 (Section B)		Duration:2 Hrs.
Examination: ESE Examination		
Programme code: 03	Class: SY	Semester: _III_(SVU 2020)
Programme: BTech EXTC	BTech EXTC	
Institute/School/Department: K. J. Somaiya School of Engineering		Name of the department: EXTC
Course Code: 116U03C303	Name of the Course: Digital Logic Design (PWD Section B)	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	List the types/ modes of operation of shift registers. What are the applications of shift register?	5
ii)	Differentiate between PROM, PAL and PLA	5
iii)	Compare synchronous and asynchronous counter	5
iv)	List and explain any 2 characteristics of Logic families	5
v)	List different techniques of analog to digital conversion. Write short note on Dual slope ADC	5
vi)	Realize using multiplexer $F(ABC) = \sum m(0,1,3,4,6)$	5

Que. No.	Question	Max. Marks
Q2	Solve the following (Any 4)	20
i)	Draw and explain block diagram of Moore and Melay machine	5
ii)	Design a mod 7 asynchronous counter using Toggle flip flops	5
iii)	If Mod 10 synchronous counter is to be designed using JK flip flops a) How many flip flops are needed? b) Write Present State Next State table for the design referring to excitation table of JK flip flop.	5
iv)	Draw 3 bit left shift register using D flip flops	5
v)	What is master Slave flip flop? Explain	5
vi)	What is state diagram and Timing diagram for a circuit? Explain with a example.	5

Que. No.	Questions	Max. Marks
Q3	Solve following Multiple choice Questions	10
i)	The Race around condition is an issue with A. T flip flop B. Master slave Flip flop	1

ii)	For the NAND Latch shown below, which of the SR input condition is invalid or forbidden ?  A. SR=00 B. SR=10 C. SR=11 D. SR=01	1
iii)	A divide by 8 Frequency divisor can be generated using A. Mod 4 counter B. Mod 8 counter C. Mod 25 counter D. 2 bit shift register	1
iv)	If 5 flip flops are connected as asynchronous counter, how many states can the counter count? A. 5 B. 10 C. 32 D. 25	1
v)	Number of states for a 3 bit twisted Ring counter A. 3 B. 4 C. 6 D. 8	1
vi)	When a flip-flop is reset, its output will be A. $Q=0, \bar{Q}=1$ B. $Q=1, \bar{Q}=0$ C. $Q=0, \bar{Q}=0$ D. $Q=1, \bar{Q}=1$	1
vii)	In a toggle mode, a JK flip-flop has A. $J=0, K=0$ B. $J=0, K=1$ C. $J=1, K=0$ D. $J=1, K=1$	1
viii)	In case of a master slave flip flop Which one of the following is the most correct? A. Master is level triggered and slave is edge-triggered B. Master is edge triggered and slave is level triggered C. Master is positive edge triggered and slave is negative edge triggered D. Master is negative edge triggered and slave is positive edge triggered	1
ix)	A basic S-R flip-flop can be constructed by cross-coupling of which basic logic gates? A. AND or OR gates B. XOR or XNOR gates C. NOR or NAND gates D. AND or NOR gates	1
x)	A square wave with a period of $10 \mu s$ drives a T flip-flop. The period of the output signal will be A. $100 \mu s$ B. $20 \mu s$ C. $10 ms$ D. $5 ms$	1



Semester: July 2024 –November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03 Programme: B.Tech	Class:SY	Semester: III(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department:EXTC	
Course Code: 116U03C304	Name of the Course: : Microprocessor and Microcontroller	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Explain the interrupt structure in 8086	5
ii)	Explain with diagram, the Minimum mode of 8086	5
iii)	Explain the execution of the following Instructions:	5
iv)	For 8086: a) MOVS B b) RCL AL, 01	5
v)	For 8051: c) MOV A, 30H d) DJNZ R1, Label e) ADD A, @R0	5
vi)	Draw and explain PSW register of 8051.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	What is register SBUF used for? Draw and explain SBUF register.	5
ii)	Explain memory organization in 8051.	5
OR		
Q2 A	What is the use of TMOD and TCON registers in 8051? Draw and explain TCON and TMOD registers.	10
Q 2 B	Solve any One	10
i)	Write an 8051 based program to Add the block of 8 bytes stored from memory location 35H. Store the result in registers R4 and R5. Also write the comments.	10
ii)	Write 8086 based assembly language program, to find odd and even bytes from the block of 10 bytes from the memory location 34200H. Store the count for odd bytes in register DL and for even bytes in DH. Also write the comments.	10

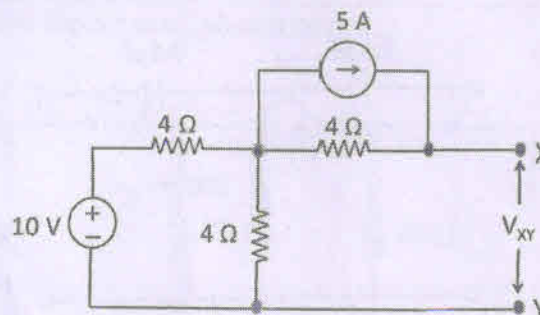
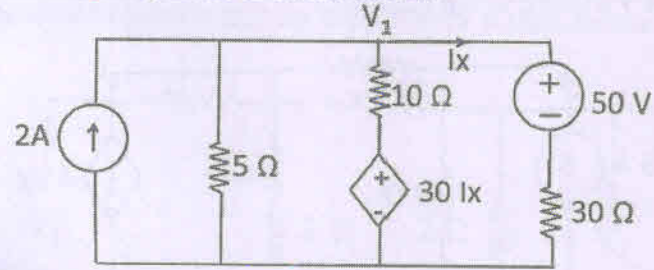
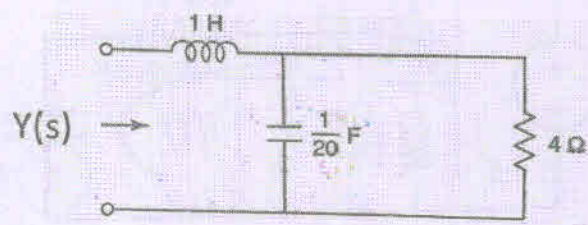
Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	How is timer used to generate the delays ? Determine the value of the initial count for timer to generate the square wave of 1 KHz. Assume the crystal freq = 12 MHz	10
ii)	Explain the addressing modes of 8051 and give an example of the each addressing mode	10
iii)	How does memory banking speed up 16 bit data transfer ? Explain the concept of memory banking in 8086.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	How to interface 14-pin LCD with 8051 ? Explain in detail using interfacing diagram	10
ii)	Explain the working of 4X4 matrix keyboard. Explain various commands in the interfacing of keyboard with 8051.	10
iii)	Explain the interfacing of ADC with 8051 alongwith the communication between the two and the control signals.	10

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Write a note on memory segmentation in 8086.	5
ii)	How is physical address of the memory location generated ? If content of the segment registers are : CS=2000H ,DS=4000H, ES =5000H, SI = 1200H, DI = 2300H. Determine the physical address of the memory location in the following instruction. i) MOV CL,[DI] ii) MOV BL,[SI]	5
iii)	Explain SHIFT and ROTATE instructions in 8086	5
iv)	List the six Interrupts in 8051 and state their use.	5
v)	Discuss the modes in Timers of 8051.	5
vi)	Explain the resolution of ADC .How does the number of bits of ADC affect it ?	5

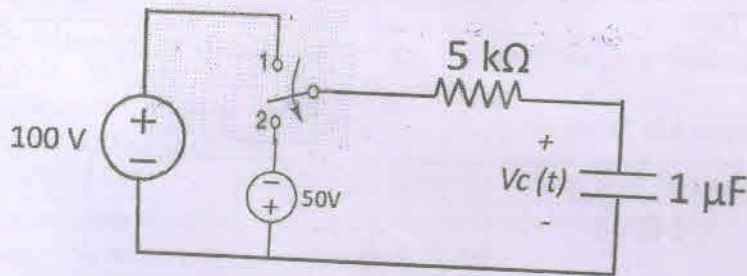


Semester: July 2024 – December 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03	Class: SY	Semester: III (SVU 2020)
Programme: B.Tech EXTC ENGG		
Name of the Constituent College: K. J. Somaiya School of Engineering		Name of the department: EXTC
Course Code: 116U03C305	Name of the Course: Electrical Network Theory	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Find V_{XY} in the following network using Superposition theorem. 	5
ii)	Determine the voltage V_1 in the network below. 	5
iii)	Find the maximum power delivered to the load if Thevenin's voltage of the circuit is 100V and equivalent resistance of the circuit is 25 Ω.	5
iv)	Check whether $P(s) = 2s^4 + 5s^3 + 6s^2 + 3s + 1$ is Hurwitz.	5
v)	List the properties of positive real function.	5
vi)	Determine $Y(s)$ for the network below 	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Find I in the following circuit.	5
The circuit diagram shows a 20V DC voltage source in series with a 5Ω resistor. The voltage across the 5Ω resistor is labeled V₁. This is followed by a dependent voltage source labeled V₂ with its positive terminal on the left. To the right of V₂ is a 4Ω resistor. After the 4Ω resistor, the circuit splits into two parallel branches. The first branch contains a 1Ω resistor in series with a dependent voltage source labeled 2V₁ (positive terminal at the top). The second branch contains a current I flowing downwards. Both branches recombine at the bottom wire.		
ii)	Find the characteristic impedance, cut-off frequency and pass band for the constant k-high pass filter shown below.	5
The circuit diagram represents a constant k-high pass filter. It consists of two series capacitors, each with a value of 0.4 μF, connected in the top line. A shunt inductor with a value of 100 mH is connected between the top and bottom lines at the junction between the two capacitors.		
OR		
Q2 A	Find the voltage V _x using nodal analysis	10
The circuit diagram for nodal analysis shows a 5A DC current source pointing upwards in parallel with a 4Ω resistor. This combination is in series with a 5Ω resistor. The voltage across the 5Ω resistor is labeled V_x with the positive terminal on the left. This is followed by a 6Ω resistor in parallel with a dependent current source labeled 4V_x pointing downwards.		
Q2 B	Solve any One	10
i)	In the network shown below, the switch is changed from position 1 to the position 2 at t = 0, steady condition having reached before switching. Find the values of i, $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at t = 0 ⁺ .	10
The circuit diagram shows a 20V DC voltage source in series with a switch. The switch can connect to position 1, which is in series with a 1H inductor, or to position 2, which is in series with a 20Ω resistor. The output terminals are connected to a 1 μF capacitor. The current through the capacitor is labeled i(t).		

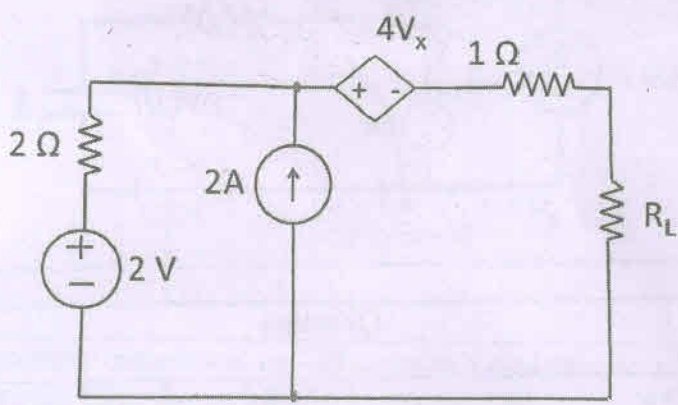
ii)	In the network shown below, the switch was in position '1' for long time. At $t=0$, the switch is moved from '1' to '2'. Find current $V_c(t)$.	10
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Find Z parameters for the network shown below.	10

ii)	Determine transmission parameters for the network shown below.	10
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iii)	Find voltage V_x in the following network using Mesh or Nodal Analysis.	10
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Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Obtain Thevenin's equivalent network for the load R_L in the network below.	10
		
ii)	Realize Foster I and Foster II form of the following LC impedance function	10
iii)	Test whether $f(s)$ is positive real function. $f(s) = \frac{s(s+3)(s+5)}{(s+1)(s+4)}$	10

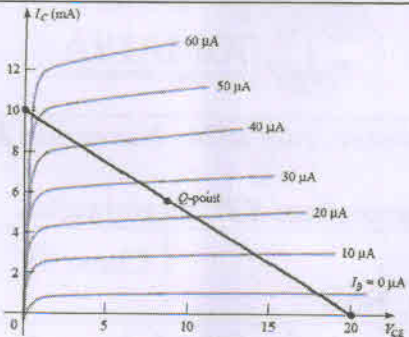
Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	In series RLC circuit, if $R = 5\ \Omega$, $L = 1\ \text{H}$ and $C = 0.5\ \text{F}$, check whether the circuit is overdamped, critically damped or underdamped.	5
ii)	State and prove maximum power transfer theorem.	5
iii)	Design constant-k low pass T section filter having cut-off frequency of 1 KHz and a nominal characteristic impedance of $300\ \Omega$.	5
iv)	Write equations of ABCD parameters and its symmetry and reciprocity conditions.	5
v)	List the properties of LC function.	5
vi)	What are initial and final conditions for the resistor, inductor and capacitor?	5

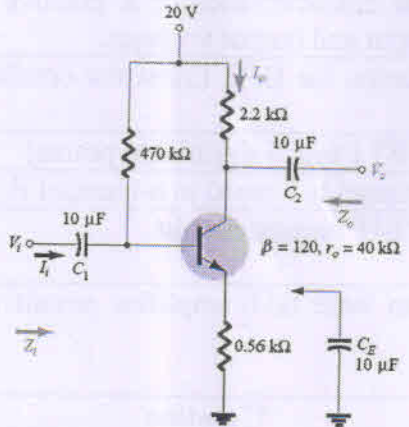
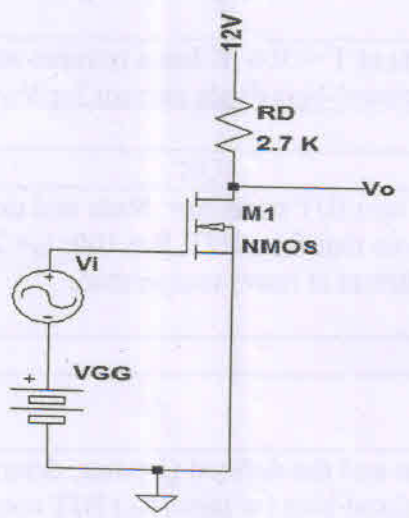


30.11.2024 (E)

Semester: July 2024 –November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03	Class: SY	Semester: III (SVU-2020)
Programme: BTECH. EXTC ENGG.		
Institute/School/Department: K. J. Somaiya School of Engineering		Name of the department: EXTC
Course Code: 116U03C302	Name of the Course: Basic Electronic Circuits	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Draw the voltage-transfer characteristics of a positive biased positive clipper. State the conditions for input and output voltages.	5
ii)	State the regions of operation for BJT. Enlist the conditions in terms of voltage and current for the same.	5
iii)	Compare BJT and MOSFET (min.5 significant points)	5
iv)	Explain in brief how a channel is formed in n-channel E-MOSFET.	5
v)	Discuss any two types of BJT biasing circuits.	5
vi)	Sketch a simple common base BJT amplifier circuit and draw it's hybrid-pi equivalent.	5
Que. No.	Question	Max. Marks
Q2	Solve the following	10
A		
i)	Determine I_{DQ} if $K_n = 0.24 \text{ mA/V}^2$, $V_{TN} = 3 \text{ V}$, $V_{GSQ} = 6.4 \text{ V}$	5
ii)	A silicon pn junction diode at $T = 300^\circ \text{K}$ has a reverse-saturation current of $I_s = 10^{-14} \text{ A}$. Determine the forward-bias diode current for $V_D = 0.5 \text{ V}$.	5
	OR	
Q2	Draw hybrid-pi model of npn BJT transistor. State and explain the significance of hybrid-pi parameters. Given that for a BJT, $\beta = 100$, $I_B = 200 \mu\text{A}$, $V_A = 80 \text{ V}$. Calculate hybrid-pi parameters at room temperature.	10
A		
Q 2	Solve any One	10
B		
i)	For the given the load line and the defined Q-point, determine the required values of V_{CC} , R_C , and R_B for a fixed-bias (without R_E) BJT configuration.	10

		
ii)	Derive the expressions for I_D , V_{GS} and V_{DS} for a voltage divider bias n-channel JFET.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	For the given circuit, determine A_v , Z_i and Z_o .	10
		
ii)	For the n-channel E-MOSFET circuit, Calculate g_m , r_d , A_v . Given that $V_{GSQ} = 3$ V, $V_{TN} = 1$ V, $K_n = 0.8$ mA/V ² , $\lambda = 0.018$ V ⁻¹ .	10
		
iii)	Explain the graphical method to determine the Q-point for fixed bias and self-bias JFET.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Compare CE, CB and CC BJT configurations.(min. 5 points)	10
ii)	State and explain the significance of stability factor of BJT. Hence derive an expression for stability factor (S_{β}) for a fixed bias circuit.	10
iii)	Explain the significance of following terms with respect to MOSFET amplifier: Transconductance (g_m), Channel length modulation parameter (λ), Pinch-off voltage(V_p), Threshold voltage(V_T), Body effect parameter (γ)	10

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Explain how heat sink is used to overcome thermal runaway in BJT.	5
ii)	Explain the working operation of varactor diode.	5
iii)	Draw DC load line of common emitter circuit and explain the significance.	5
iv)	What do you mean by Early effect?	5
v)	State and explain the relationship for drain current and gate-source voltage for MOSFET in non-saturation region.	5
vi)	What is the role of capacitor in clamper circuit?	5